(different numbers of particle families, different number and types of fundamental forces, among other substantial deviations), a few entries in the catalog yield physics that is qualitatively close to what we actually observe. That is, there are examples of Calabi-Yau spaces that, when chosen for the curled-up dimensions required by string theory, give rise to string vibrations that are closely akin to the particles of the standard model. And, of prime importance, string theory successfully stitches the gravitational force into this quantum-mechanical framework.

With our present level of understanding, this situation is the best we could have hoped for. If many of the Calabi-Yau shapes were in rough agreement with experiment, the link between a specific choice and the physics we observe would be less compelling. Many choices could fit the bill and hence none would appear to be singled out, even from an experimental perspective. On the other hand, if none of the Calabi-Yau shapes came even remotely close to yielding observed physical properties, it would seem that string theory; although a beautiful theoretical framework, could have no relevance for our universe. Finding a small number of Calabi-Yau shapes that, with our present, fairly coarse ability to determine detailed physical implications, appear to be well within the ballpark of acceptability is an extremely encouraging outcome.

Explaining the elementary matter and force particle properties would be among the greatest—if not the greatest—of scientific achievements. Nevertheless, you might ask whether there are any string theoretic predictions—as opposed to postdictions—that experimental physicists could attempt to confirm, either now or in the foreseeable future. There are.

Superparticles

The theoretical hurdles currently preventing us from extracting detailed string predictions force us to search for generic, rather than specific, aspects of a universe consisting of strings. Generic in this context refers to characteristics that are so fundamental to string theory that they are fairly insensitive to, if not completely independent of, those detailed properties of the theory that are now beyond our theoretical purview. Such characteristics can be discussed with confidence, even with an incomplete understanding of the full theory